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Individuals with higher risk perception report more heat stress and thirst after hiking in a hot and dry environment without other differences in physiological measurements

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Supplementary material for this article is available online

Abstract

This study investigated the relation between physiological markers associated with exercise (measured and self-reported) and self-reported risk perception (RP) among hikers in extremely hot environments. Participants ($n = 340$, 43% female, 40.1 $\pm$ 12.9 yr) were recruited at hiking trails during two hot and dry summers in Phoenix, Arizona, USA. Participants completed an onsite exit questionnaire, and a subsample participated in physiological testing. Outcomes were stratified for participants with a high and low RP. Self-reported heat stress (3.0, interquartile range (IQR): 3.0–5.3 vs. 3.0, IQR: 2.0–4.0, $p = 0.001$) and thirst (38.0, IQR: 23.0–52.0 vs. 31.0, IQR: 19.0–47.8, $p = 0.039$) were significantly higher in hikers with high RP vs. low RP scores, respectively. However, perceived exertion (6.0, IQR: 6.0–12.0), body temperature (35.3 °C, IQR: 34.2–36.0), urine specific gravity (1.021, IQR: 1.015–1.026), and fitness level (48.6, IQR: 41.9–56.7), were not associated with RP scores ($p \geqslant 0.100$). Most participants considered ambient temperature and hike duration when planning fluid intake, however, although the high-risk group reported these safety behaviors more frequently, no group difference was seen in hydration status, amount of fluid carried (1000 ml, IQR: 500–1999), or fluid consumed (591 ml, IQR: 355–1000). In conclusion, RP may influence how individuals anticipate and respond to environmental challenges, such as reporting more heat stress and thirst in those with a high RP. Greater awareness regarding a heat safety campaign and precautionary behaviors were seen in high RP, but this did not translate into measurable differences in other physiological outcomes or actual hydration practices.

1. Introduction

Outdoor recreation activities such as hiking are driven by a sense of self-development, freedom, control, or thrill, but also pose inherent risks [1]. One such risk is heat injury. Over 200 hikers are rescued every year due to heat injury or dehydration within the Phoenix, Arizona (AZ) hiking trail system [2]. To mitigate heat injury risk, the City of Phoenix launched the 'Take a Hike, Do it Right' campaign in 2015. The campaign educates the public about the risks and signs of exertional heat illness and ways to mitigate its onset. Although much is known about the physiology of heat stress and dehydration [3, 4], little is known about how these processes affect risk perception (RP) and the associated risk-taking behaviors, especially among hikers during strenuous hikes in hot environments. The elements of self-reported RP and risk-taking are multifaceted decision-making processes [5] influenced by the type of activity being engaged, environmental conditions, and the individual's fitness and adaptation level [3, 4, 6].

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Understanding the relationship between physiological responses to extreme heat and RP is therefore imperative to reduce risk behavior in hikers.

The human body is tightly controlled to maintain a narrow core body temperature range of 36 °C–38 °C, which can easily increase to 40 °C or beyond during exercise or in hot environments [7]. When the internal body temperature rises above 40 °C, there is an increased risk of heat injury/illness, which is higher when exercising in hot environments in a dehydrated state [8–10]. In daily life, continued exposure to heat stress leads to multiple metabolic alterations to better handle the heat, i.e. metabolic and physiological adaptations that allow for better fluid balance, increased cardiovascular stability, improved skeletal muscle metabolism, and increased heat tolerance [11, 12]. However, without proper hydration or cool-seeking behaviors, heat illness may occur [8–10]. Exercise combined with heat stress leads to exertional heat illness that is often characterized by elevated core body temperature, muscle cramping, heat exhaustion, or heatstroke [13].

While the effects of heat stress and dehydration on physiological function are widely studied [7], the interaction between these physiological states and RP and/or risk behavior during hiking exercise is not well documented. Heat stress can impact the ability to perform both simple and complex tasks [14, 15], and acute heat exposure in an occupational work environment can decrease RP with increased risk-taking behaviors as a result [16]. Therefore, elevated ambient temperatures may increase the prevalence of unsafe behaviors [17]. Although mild to moderate dehydration has been shown to produce decrements in aspects of cognitive function such as perceptual discrimination, visuomotor tracking, and psychomotor skills [18, 19], other studies show inconsistent results [20, 21]. Within the scope of dehydration studies, research mainly focuses on the effects of fluid on cognitive processes or performance, but no specific research has investigated the relationship between markers of hydration and RP in hikers.

Field studies in parks and recreation management have explored RP and risk behavior in relation to hiking, identifying the components of risk as physical, social, and psychological, classifying RP into subjective and objective categories and uncovering the prevention of risk behavior [3, 6, 22, 23]. Although the fields of health and recreation have investigated hiker safety from a physiological and psychosocial perspective respectively, there is a relative paucity in research integrating the interaction of these two distinct domains. For example, studies looked at different areas of experienced risk in hiking, for example identifying, physical and psychological dimensions of perceived risk in hiking without integrating physiological data to assess its relationship with RP [23]. Or they defined personal risk in outdoor recreation, but without reporting any objective physiological outcomes related to risk measures [24, 25]. Finally, investigators have looked at the physiological impact of trails (as a result of trail difficulty), but in that case not addressing RP [26].

Therefore, this study integrates physiological and psychosocial data to offer a more comprehensive understanding of risk behaviors and perceptions in extreme heat conditions. Specifically, we examine the relationship between RP, subjective measures (such as, perceived exertion, heat stress, and thirst), and objective physiological parameters (including core body temperature, hydration status, and cardiorespiratory fitness). This interdisciplinary approach aims to elucidate how physiological and perceptual factors interact to influence hiker safety under thermal stress.

Based on previous literature, the following hypotheses were tested while categorizing participants accordingly to low vs. high RP scores:

H1: Self-reported perceived exertion, heat stress, and thirst after a hike in hot environmental conditions are higher in hikers with high RP scores.

H2: After a hike in hot environmental conditions, indirect markers of heat stress, such as high core body temperature, a high urine concentration, and a low fitness level are positively associated with higher RP scores.

H3: Familiarity with heat risk and mitigation strategies, including the degree of preparation for the heat, while using hydration strategies, is higher in hikers with high RP scores.

2. Methods and materials

During the summer of 2021, data was collected at the Piestewa Peak Summit Trail located in the City of Phoenix, AZ. This mountain trail is rated as extremely difficult by the City's Parks and Recreation Department and is very popular among city dwellers, for example to train for a hike in the Grand Canyon National Park. During the summer of 2022, data was collected at a lower profile mountain trail close to a university setting. This trail is rated easy and is popular amongst regular hikers. During the first year, trained researchers systematically and randomly approached potential participants post-hike from July to October 2021. Sampling occurred mostly on weekends and in the morning to mimic typical

summer use. During the second year, the study was advertised using social media ads, and targeted participants who were less active compared to the previous sample. Specifically, participants who exercised once a week or less were sought to ensure a wider range of fitness levels was represented in the cohort. Sampling occurred in June and July of 2022. Data collection for both years was performed at the end of the hike.

Regardless of data collection year, the data collection was divided into two parts: first, upon agreeing to participate in the study, participant's heart rate was measured, and a thermal image was taken to allow participant classification based on their body temperature. Participants then completed a 10–15 min on-site, self-administered exit questionnaire. The second part included two additional physiological measurements involving taking a urine sample and performing an exercise step test. Participants could opt out of the additional element of data collection at any point.

2.1. Participants

The study included all interested participants who were at least 18 years of age and who hiked for 30 min or more. During the second year, participants who exercised once per week or less were recruited to offset the high fitness level of the first-year cohort. A minimum of 289 participants was needed to achieve a confidence level of 95% with a margin of error of 5%. This study was approved by the Institutional Review Board at AZ State University (STUDY00014107), and informed consent was included.

2.2. Procedures

Upon completing their hike, the selected participants signed the informed consent and were instructed to refrain from drinking or eating anything during Part I of the measurements (∼10 min) so as not to interfere with the subjective measurements. All participants were assigned a subject ID to ensure anonymity within the data set.

Part I: A headshot was taken with a thermal camera to record body temperature immediately after each participant agreed to their involvement in the study by signing consent. Participants were then seated to complete a 10 min online questionnaire via their mobile phone or staff-supplied tablet while a pulse oximeter simultaneously recorded the heart rate. During this time, a researcher analyzed the size and contents of the fluid container(s) hikers brought with them to the trail, allowing for an estimation of the volume of fluid carried and consumed during the hike.

Part II: For those who agreed, a urine sample was collected to assess hydration status. Additionally, a modified YMCA step test [27], was performed to determine cardiorespiratory fitness. Finally, during each data collection day ambient environmental conditions such as ambient temperature and humidity were registered.

2.3. Measurements

Questionnaire: A 30-question online questionnaire (supplementary file 1) was used to collect self-reported data. The questions were categorized as follows: personal demographics, RP, scales reflecting self-reported physiological outcomes, and familiarity with heat risk and mitigation strategies.

Personal demographics and hiking duration (#10 questions): Body height (reported as feet and inches, converted to centimeters), body mass (reported as pounds, converted to kg), age (years), sex/gender, ethnicity, race, employment status, highest education level, income, and duration of the hike (minutes).

RP (#2 questions): The combined RP score utilized in our study was adapted from two established instruments described by Kalkstein & Sheridan in which participants were asked to rate how dangerous they perceived the Phoenix heat to be [28], as well as, She *et al* in which participants were asked to assess concerns related to hiking [23]. These scales have been previously employed in studies assessing RP in similar contexts. This resulted in one question about how dangerous the Phoenix heat is perceived to be by the participants (not at all dangerous to extremely dangerous, scale 1–5) [28], and 15 questions about worries during hiking related to RP, such as worry about accidents, slips or falling during the hike (15 elements: strongly disagree to strongly agree, scale 1–5) [23].

Rate of perceived exertion, heat stress, and thirst scales (#3 questions): Rate of perceived exertion (Borg scale 6–20) [29], heat stress scale (very mild to excruciating unbearable, scale 0–10) [30], and thirst scale (not thirsty at all to extremely thirsty, scale 0–100) [31].

Familiarity with heat risk and mitigation strategies (#5 questions): Awareness of the Phoenix heat warning system (yes/no); awareness of the City of Phoenix 'Take a Hike, Do It Right' campaign (yes/no) and if yes, familiarity with this campaign and its pre-hiking safety strategies (eight elements, scale 1–5) [2].

Further, participants were asked how they prepared for the heat (eight elements, check all that apply). This was followed by two questions about fluid, including factors taken into consideration while deciding how much fluid to bring (nine elements, check all that apply), and how satisfied participants were with the amount of fluid brought (very dissatisfied to very satisfied, scale 1–5).

Physiological markers:

Thermal Imaging as a marker for body temperature: A handheld infrared camera image (FLIR E95, Teledyne FLIR, Wilson, Oregon, USA) was taken 30–40 cm from the forehead while the subject was seated on a stool in front of a grey brick wall. The forehead is considered to reasonably reflect core body temperature if adjusting for the fact that the skin temperature at that point is around 1 °C lower than the actual body core temperature [32], as no further temperature adjustments were made the variable will be expressed as 'body temperature'. Participants were brought indoors, with a stable ambient temperature of ∼26 °C, with emissivity set at 0.98 for human skin, to take a picture, and then brought back outside. Participants were photographed within 30–60 s of entering the room. A heat map of the headshots was generated using the free FLIR thermal studio starter program. Environmental ambient temperature and humidity for measurements were inserted in the program parameters, and a 3 × 3 pixel area of the internal canthus of the eye was selected [33] to retrieve the average, minimum, and maximum body temperature.

Hydration status: Urine was collected in a 32 fl oz (∼990 ml) urine container. The hydration status was quantified by analyzing the urine sample via urine specific gravity refractometry (Atago 4410 PAL-10S Digital Handheld Pocket Urine S.G. Refractometer, 1–1.06 urine specific gravity range, Bellevue, Washington). Each sample was measured twice, if the measurements differed by more than 0.005, a third measurement was taken, and the median value was accepted.

Fluid brought and fluid consumed: All fluid containers brought on the hike were inspected by a research team member, and the volume of the container was estimated using a predetermined list of the most frequent sizes of bottles, containers, and water bags. Then, the research team member inquired how much fluid was available in each container before the start of the hike to estimate the total volume consumed by subtracting the remaining fluid.

Cardiorespiratory fitness: VO_2max was assessed by using a three-minute modified YMCA step test protocol. A 12-inch (30.5 cm) step height was used for all participants regardless of sex, and a 96 beats-per-minute cadence was adopted for the test (resulting in 24 cycles of steps completed per minute). For pragmatic reasons, the step height was not differentiated for men and women, as indicated in the original protocol [27]. Heart rate was measured before and during the test and for one minute after the test to allow calculating heart rate recovery using a pulse oximeter (Zacurate 500BL Fingertip Pulse Oximeter Blood Saturation Monitor, Stafford, Texas). Based on the maximal heart rate registered, VO_2max was estimated using the following sex specific equations [27]:

Males: $VO_2\text{max} = 70.597–0.246 \times (\text{Age}) + 0.077 \times (\text{Height})—0.222 \times (\text{Weight})—0.147 \times (\text{HR})$, and Females: $VO_2\text{max} = 70.597–0.185 \times (\text{Age}) + 0.097 \times (\text{Height})—0.246 \times (\text{Weight})—0.122 \times (\text{HR})$.

Environmental conditions: Ambient temperature and humidity were recorded during each trial period using a Kestrel 5400 Heat Stress Tracker (Nielsen–Kellerman, Boothwyn, PA, USA).

2.4. Calculating RP scores and stratifications

To calculate total heat-related RP scores, the outcomes of the following two questions were combined: (1) 'how dangerous you feel the Phoenix heat is for you.' Subjects answered on a five-point Likert scale (1–5, not at all dangerous to extremely dangerous), for five possible points. (2)'When hiking, I worry about…' they responded on a five-point Likert scale (1–5, not at all worried to extremely worried) and with 15 separate elements (removing the elements on infectious disease and contaminated foods) for 75 possible points. Combining the two questions resulted in a minimum score of 16 and a maximum score of 80, where higher scores indicate a higher RP (supplementary file 1, questions 16 and 17).

Median RP scores were used to split participants into a high-RP (HRP) and a low-RP (LRP) group. The thermal imaging body temperature was stratified based on group median. Self-reported rates of perceived exertion, heat stress, and thirst perception were stratified using the group median. Additionally, body temperature was split into low vs. high values based on the median. Hydration levels were categorized as euhydrated (USG < 1.020) and underhydrated (USG $\geqslant 1.020$). Fluid carried and consumed were stratified into low vs. high groups based on the median. Finally, participants were categorized by fitness level based on the median-estimated VO_2max values (low vs. high).

2.5. Data analysis

SPSS version 25 and excel were used for statistical analyses. Data was presented as mean ± standard deviation (SD), median, interquartile range (IQR), or as frequencies (n) and percentages (%). When relevant, Cronbach's alpha (α) was calculated to assess the internal consistency of a set of questions. Data was stratified to allow for comparison between groups with high and low risk behavior. Specifically, we tested for any differences in physiological metrics between HRP and LRP groups. Mann–Whitney U tests compared results for continuous variables and Likert scales. Rank biserial correlation effect size (r_{rb})- with small (∼0.1), medium (∼.3), and large (⩾0.5) effect sizes- was calculated for differences in self-reported and physiological metrics. Categorical questions were assessed with Pearson chi-squared tests or Fisher's exact tests (with Cramer's V effect size), with small (⩽0.1), medium (0.1–0.3), and large (>0.3) effect sizes. In addition, RP, physiological metrics, and self-reported outcomes were correlated against each other using Spearman correlations. All significance levels were set at $p \leqslant 0.05$.

3. Results

Out of 901 hikers that were approached, 340 (37.7%) agreed to participate in the study. A total of $n = 319$ participants were recruited in year 1 on the extremely difficult trail, and 21 hikers were recruited in year 2 on the easy trail. As no clear differences were seen in outcomes between groups, they were combined ($p > 0.05$). Height, weight, BMI, and age were normally distributed. Participants (of which 42.9% were female) reported an average age of 40.6 ± 12.9 yr (table 1). The median hiking duration of the participants was 90 min (IQR: 60–120 min) without a significant difference between RP groups ($p \geqslant 0.71$). The average ambient temperature was 31.4 ± 2.9 °C with 46.9 ± 12.4% relative humidity, without a significant difference between RP groups ($p \geqslant 0.24$). A total of 73.1% ($n = 249$) of the participants had no complaints during their hike. The most common complaints of the remaining 26.9% ($n = 91$) included being light-headed or dizzy (9.4%, $n = 32$), or having muscle cramps (5.1%, $n = 17$) or the urge to urinate (4.4%, $n = 15$).

3.1. RP

The median RP score for the sample was 48.0 (40.3–54.0). The internal consistency for RP was high with Cronbach's α (α) = 0.88. When stratifying for RP, the LRP (LRP, $n = 184$) group median score was 42.0 (36.0–46.0), and the HRP (HRP, $n = 156$) group median score was 55.0 (51.0–59.0). As shown in table 2, hikers worry the most about accidents (4.0 [3.0–4.0]), followed by inadequate peers to hike with (3.0 [2.3–4.0]), and after this they worry equally about multiple items, including physical exhaustion, lack of shade, and not having medical assistance available when in need (3.0 [2.0–4.0]). After stratifying the group for low RP vs. high RP, all individual RP elements were reported higher in the HRP group ($p < 0.001$), with moderate to large effect sizes from $r_{rb} = 0.38$ to $r_{rb} = 0.59$. The ratio difference for sex in low vs. high RP groups was driven by lower scores in males: 47.0 (40.0–52.0) vs. females 50.0 (43.0–56.3), $p = 0.01$.

3.2. Self-reported physiological outcomes

As shown in table 3, significant differences existed between the low and high RP groups for self-reported heat stress (3.0 [2.0–4.0] vs. 3.0 [3.0–5.3], respectively, $p = 0.001$, with small to medium effect size $r_{rb} = -0.20$), and for thirst (31.0 [19.0–47.8] vs. 38.0 [23.0–52.0], respectively, $p = 0.039$, with small effect size $r_{rb} = -0.15$). However, there were no differences between the two groups for the rate of perceived exertion (6.0 [6.0–12.0], $p = 0.10$). No sex differences were reported for any of the outcomes reported, $p \geqslant 0.29$.

3.3. Physiological metrics

No significant difference among the RP groups was found for directly measured physiological metrics, as indicated in table 3. The two RP groups did not differ in body temperature (35.3 [34.2–36.0]), hydration outcomes, (1.021 [1.015–1.026]), fluid brought (1000 [500–1999]), fluid consumed (591 [355–1000]), or fitness level (48.6 [41.9–56.7]), $p > 0.05$. Fitness level was based on the outcome of the step test, and heart rate at baseline was 83.0 bpm (76.0–93.0), and after the step test it peaked at 131 bpm (120–145), following a heart rate recovery after one minute of 38.0 bpm (32.0–47.0), without any difference between RP groups, $p \geqslant 0.16$. No sex differences were reported for any of the outcomes reported ($p \geqslant 0.11$), except for the estimated VO_2max (males: 55.5 ml kg^{-1} [47.5–59.1] vs. females: 42.4 ml kg^{-1} [39.2–45.9], $p < 0.001$).

Table 1. Demographics (mean ± sd, or % and *n*) of participants and stratified for high and low risk perception, and *p*-value for significance ($n = 340$).

Descriptives	Total group ($n = 340$)	Low-risk perception (LRP, $n = 184$)	High-risk perception (HRP, $n = 156$)	*p*-value
Body height (cm)	173 (165–180)	173 (168–180)	170 (165–178)	0.067
Body mass (kg)	72.6 (63.5–83.9)	74.4 (63.5–83.9)	72.6 (61.4–83.8)	0.423
BMI	24.1 (22.2–27.3)	23.9 (22.3–27.2)	24.3 (22.0–27.3)	0.770
Age (years)	41.0 (30.0–51.0)	42.0 (29.3–52.0)	40.0 (30.0–50.0)	0.299
Sex/gender				
Male	54.4% (185)	60.3% (111)	47.4% (74)	0.015
Female	42.9% (146)	35.9% (66)	51.3% (80)	
Non-binary/third gender	0.9% (3)	0.5% (1)	1.3% (2)	
Prefer not to say	0.3% (1)	0.5% (1)	0% (0)	
Missing	1.5% (5)	2.7% (5)	0% (0)	
Ethnicity				
Hispanic or Latino	20.6% (70)	20.1% (37)	21.2% (33)	0.866
Race				
American Indian/Alaskan Native	2.9% (10)	2.2% (4)	3.8% (6)	0.351
Asian	10.9% (37)	9.8% (18)	12.2% (19)	0.465
Black or African American	4.1% (14)	4.9% (9)	3.2% (5)	0.452
Native Hawaiian/Pacific Islander	1.2% (4)	1.1% (2)	1.3% (2)	0.857
White	80.0% (272)	81.0% (149)	78.8% (123)	0.201
Employment status				
Employed for wages	68.8% (234)	67.9% (125)	69.9% (109)	0.546
Self employed	10.6% (36)	9.8% (18)	11.5% (18)	
Out of work, looking	2.6% (9)	2.7% (5)	2.6% (4)	
Out of work, not looking	1.2% (4)	1.1% (2)	1.3% (2)	
Student	7.9% (27)	6.5% (12)	9.6% (15)	
Homemaker	1.8% (6)	1.1% (2)	2.6% (4)	
Retired	3.2% (11)	5.4% (10)	0.6% (1)	
Military	0.9% (3)	0.5% (1)	1.3% (2)	
Unable to work	0.9% (3)	1.1% (2)	0.6% (1)	
Missing	2.1% (7)	3.8% (7)	0% (0)	
Education				
Some high school	1.5% (5)	1.1% (2)	1.9% (3)	0.241
High school graduate	6.8% (23)	9.2% (17)	3.8% (6)	
Vocational/trade school	2.9% (10)	3.3% (6)	2.6% (4)	
Some college	13.8% (47)	14.7% (27)	12.8% (20)	
Two year college degree	9.1% (31)	8.7% (16)	9.6% (15)	
Four year college degree	36.2% (123)	33.2% (61)	39.7% (62)	
Master's degree	21.2% (72)	18.5% (34)	24.4% (38)	
Ph.D., M.D., J.D., or equiv.	7.4% (25)	9.2% (17)	5.1% (8)	
Missing	1.2% (4)	2.2% (4)	0% (0)	

LRP = low risk perception group, HRP = high-risk perception group.Each question was based on a Likert scale from 1 ('Strongly Disagree') to 5 ('Strongly Agree').Biserial effect sizes (r_{bs}), quantified the strength and direction of these differences. Negative effect size values suggest higher values in the high-risk group. Significance was set at $p \leqslant 0.05$.

3.4. Correlations between physiological metrics and self-reported outcomes

The strength of the relationship between outcomes, including the earlier reported significant difference for heat stress and thirst between RP groups, was tested by running Spearman correlations. RP was significantly and positively correlated with self-reported heat perception ($r(331) = 0.230$, $p < 0.01$), indicating that participants with a higher RP score were more likely to perceive more heat stress. In addition, participants' self-reported heat perception during the hike was moderately, and positively correlated with their thirst levels ($r(315) = 0.424$, $p < 0.01$), indicating that as hikers felt hotter, they reported feeling more thirsty. Further, a significant positive correlation was found between the volume of fluid brought by hikers and the volume consumed ($r(288) = 0.767$, $p < 0.01$). No other significant correlations were observed.

Table 2. Risk perception scores reported as median and interquartile range, effect size and *p*-value for significance ($n = 340$).

Questionnaire question	Total group	LRP	HRP	Effect size, *p-value*
How dangerous do you think the Phoenix heat is for you?	3.0 (3.0–4.0)	3.0 (2.0–4.0)	4.0 (3.0–4.0)	−0.46, <0.001
I worry about:				
Accidents such as falling, slips, injuries, etc.	4.0 (3.0–4.0)	3.0 (3.0–4.0)	4.0 (4.0–5.0)	−0.46, <0.001
Insufficient and inadequate peers would affect the team	3.0 (2.3–4.0)	3.0 (2.0–4.0)	4.0 (3.0–4.0)	−0.39, <0.001
Exertional heat illness (fever, nausea)	3.0 (2.0–4.0)	3.0 (2.0–3.8)	4.0 (3.0–4.0)	−0.56, <0.001
Physical exhaustion	3.0 (2.0–4.0)	3.0 (2.0–3.0)	4.0 (3.0–4.0)	−0.57, <0.001
Lack of shade	3.0 (2.0–4.0)	3.0 (2.0–4.0)	4.0 (3.0–4.0)	−0.50, <0.001
Sudden change in weather conditions	3.0 (2.0–4.0)	3.0 (2.0–3.0)	4.0 (3.0–4.0)	−0.49, <0.001
Outdoor medical assistance is not accessible when in need	3.0 (2.0–4.0)	2.0 (2.0–3.0)	4.0 (3.0–4.0)	−0.50, <0.001
Poor communication with the outside world	3.0 (2.0–4.0)	2.0 (2.0–3.0)	3.0 (3.0–4.0)	−0.51, <0.001
Wildlife invasion, insect bites, snake bites	3.0 (2.0–4.0)	3.0 (2.0–3.0)	4.0 (3.0–4.0)	−0.45, <0.001
Getting lost during the hike	2.0 (2.0–4.0)	2.0 (1.0–3.0)	3.0 (2.0–4.0)	−0.56, <0.001
Carrying insufficient fluid	3.0 (2.0–4.0)	2.0 (1.0–4.0)	4.0 (3.0–5.0)	−0.59, <0.001
Carrying insufficient snacks/ food	3.0 (2.0–4.0)	2.0 (2.0–3.0)	4.0 (3.0–4.0)	−0.59, <0.001
Loss of property during hiking	2.0 (2.0–3.0)	2.0 (1.0–2.8)	3.0 (2.0–3.0)	−0.38, <0.001
Hiking experience falling short of expectations	2.0 (2.0–3.0)	2.0 (1.0–2.0)	3.0 (2.0–3.0)	−0.43, <0.001
Feeling uncomfortable in unfamiliar surroundings	2.0 (2.0–3.0)	2.0 (1.0–2.0)	3.0 (2.0–4.0)	−0.58, <0.001

LRP = low risk perception group, HRP = high-risk perception group. Each question was based on a Likert scale from 1 ('Strongly Disagree') to 5 ('Strongly Agree'). Biserial effect sizes (r_{bs}), quantified the strength and direction of these differences. Negative effect size values suggest higher values in the high-risk group. Significance was set at $p \leqslant 0.05$.

3.5. Heat risk and mitigation strategies

A total of 76.5% ($n = 257$) of the participants were familiar with the Phoenix heat warning system, with no difference between risk groups (low RP: 77.2%, $n = 139$ vs. high RP: 75.6% ($n = 118$), with X^2 (1) = 0.12, $p = 0.733$). When asking whether they were familiar with the City of Phoenix 'Take a Hike, Do it Right' campaign, 42.9% ($n = 146$) reported to be familiar with this campaign, with the low RP group reporting a numerically lower familiarity (38.3%, $n = 69$) than the high RP group (49.4%, $n = 77$)(X^2 (1) = 4.14, $p = 0.042$).

The subgroup reporting to know about the 'Take a Hike, Do it Right' campaign was also asked how familiar they were with its recommendations toward hiker safety, as listed in table 4. The internal consistency of the reported data in the table was high (Cronbach's α of 0.980). Most of the hikers ($n = 146$) were familiar with the advice to bring water (5.0 [4.0–5.0]), followed by taking responsibility (4.0 [4.0–5.0]), and keeping in contact with others (4.0 [3.0–5.0]. The least known advice was to watch the weather (3.0 [2.0–4.0]). No significant differences were found in familiarity with the campaign between low and high RP groups ($p \geqslant 0.075$).

Preparation for the heat

When participants were asked if they applied specific clothing choices to prepare for or mitigate heat stress, they reported using a hat (54.4%, $n = 185$), sunglasses (47.9%, $n = 163$), a cool shirt (45.6%, $n = 155$), a cool vest (2.4%, $n = 8$), or clothes soaked in water (2.1%, $n = 7$), but the usage of these items did not differ between RP groups ($p \geqslant 0.116$). Only the use of a cool scarf or bandana (8.8%, $n = 30$) significantly differed between low RP (5.4%, $n = 10$) and high RP (12.8%, $n = 20$), with $p = 0.017$ and small to moderate effect size $V = 0.13$. Finally, of all participants, 14.1% ($n = 48$) reported they did not use any of these strategies.

Hydration strategies

As shown in table 5, today's temperature (65.9%, $n = 224$) and planned duration of the hike (66.2%, $n = 225$) were the most popular responses to determine how much fluid participants would bring, followed by the amount of fluid participants were able to carry (40.9%, $n = 139$). Participants in the high RP group (73.7%, $n = 115$) were more likely to consider the planned duration of the hike than low RP (59.8%, 110), with $p = 0.002$, $V = 0.17$). Additionally, participants in the high RP group (74.4%, $n = 116$) were more likely to consider the daily ambient temperature than low RP (58.7%, $n = 108$), with $p = 0.002$, $V = 0.17$. Other factors such as the amount of fluid hikers were able to carry (40.9%, $n = 139$), available bottle size (24.1%, $n = 82$), and the water/restroom facilities on the trail (16.8%, $n = 57$) were of lesser importance and did not significantly differ between RP groups ($p \geqslant 0.055$).

Table 3. Measured or self-reported physiological metrics for the total group and stratified for low and high-risk perception.

	Total group median (IQR)	Risk group (n)	Median (IQR)	Effect Size, p-value
Rate of perceived exertion (Borg Scale: 6–20, $n = 281$)	6.0 (6.0–12.0)	Low ($n = 155$)	6.0 (6.0–11.0)	−0.10, 0.100
		High ($n = 126$)	6.0 (6.0–12.0)	
Self–perceived Heat (1–10, $n = 333$)	3.0 (2.0–5.0)	Low ($n = 179$)	3.0 (2.0–4.0)	−0.20, 0.001
		High ($n = 154$)	3.0 (3.0–5.3)	
Self–perceived Thirst (1–100, $n = 319$)	34.0 (21.0–50.0)	Low ($n = 167$)	31.0 (19.0–47.8)	−0.15, 0.039
		High ($n = 150$)	38.0 (23.0–52.0)	
Body temperature (Thermal imaging, °C, $n = 116$)	35.3 (34.2–36.0)	Low ($n = 65$)	35.2 (34.5–35.9)	−0.04, 0.722
		High ($n = 51$)	35.4 (33.4–36.2)	
Hydration (USG) ($n = 126$)	1.021 (1.015–1.026)	Low ($n = 66$)	1.021 (1.014–1.026)	0.06, 0.998
		High ($n = 64$)	1.021 (1.015–1.024)	
Fluid brought (ml, $n = 290$)	1000 (500–1999)	Low ($n = 164$)	1000 (501–1999)	0.06, 0.399
		High ($n = 126$)	946 (500–1893)	
Fluid consumed (ml, $n = 290$)	591 (355–1000)	Low ($n = 164$)	600 (400–1000)	0.26, 0.556
		High ($n = 164$)	550 (336–1000)	
Estimated VO_2max (ml kg $^{-1}$min^{-1}, $n = 132$)	48.6 (41.9–56.7)	Low ($n = 66$)	53.4 (41.5–58.5)	0.16, 0.121
		High ($n = 66$)	44.9 (42.0–56.0)	

LRP = low risk perception group, HRP = high-risk perception group. Biserial effect sizes (r_{bs}), quantified the strength and direction of these differences. Negative effect size values suggest higher values in the high-risk group. Significance was set at $p \leqslant 0.05$.

Table 4. Self-reported responses to the question of how familiar participants were with elements of the 'Take a Hike, Do it Right' campaign, reported as median and interquartile range, effect size, and p-value for significance ($n = 146$).

Questionnaire question	Total group	LRP	HRP	ES, p-value
Bring water	5.0 (4.0–5.0)	5.0 (4.0–5.0)	5.0 (4.0–5.0)	−0.08, 0.327
Take responsibility	4.0 (4.0–5.0)	4.0 (3.0–5.0)	4.0 (4.0–5.0)	−0.06, 0.481
Keep in contact	4.0 (3.0–5.0)	4.0 (2.5–5.0)	4.0 (3.0–5.0)	−0.16, 0.075
Do not trailblaze	4.0 (3.0–5.0)	4.0 (3.0–5.0)	4.0 (3.0–5.0)	0.09, 0.317
Dress appropriately	4.0 (3.0–4.0)	4.0 (2.0–5.0)	4.0 (3.0–4.0)	−0.11, 0.222
Team up	4.0 (3.0–4.0)	4.0 (2.0–4.5)	4.0 (3.0–4.0)	−0.09, 0.311
Be honest about your physical and medical condition	4.0 (3.0–4.0)	4.0 (3.0–5.0)	4.0 (3.0–5.0)	−0.08, 0.389
Watch the weather	3.0 (2.0–4.0)	3.0 (2.0–4.0)	4.0 (2.0–4.0)	−0.08, 0.385

LRP = low risk perception group, HRP = high-risk perception group. Each question was based on a Likert scale from 1 ('Strongly Disagree') to 5 ('Strongly Agree'). ES = Biserial effect sizes (r_{bs}), quantified the strength and direction of these differences. Negative effect size values suggest higher values in HRP vs. LRP, indicating more pronounced perceptions or behaviors related to the risk factors assessed. Significance was set at $p \leqslant 0.05$.

Table 5. Self-reported responses about factors considered while deciding how much fluid to bring on this hike, reported as median and interquartile range, effect size, and p-value for significance ($n = 340$).

Items	Total group	LRP	HRP	Effect size, p-value
Planned duration of the hike	66.2% (225)	59.8% (110)	73.7% (115)	0.15, 0.007
Today's temperature	65.9% (224)	58.7% (108)	74.4% (116)	0.17, 0.002 [f]
The amount of fluid that I am able to carry	40.9% (139)	36.4% (67)	46.2% (72)	0.10, 0.069
Available bottle or container size	24.1% (82)	22.8% (42)	25.6% (40)	0.03, 0.545
The water/restroom facilities on this trail	16.8% (57)	16.3% (30)	17.3% (27)	0.01, 0.805
Other	5.9% (20)	6.5% (12)	5.1% (8)	0.03, 0.586
I did not really plan my fluid intake	5.0% (17)	6.0% (11)	3.8% (6)	0.05, 0.369
None	2.4% (8)	3.8% (7)	0.6% (1)	0.10, 0.055 [f]
My financial situation	1% (3)	1.6% (3)	0.0% (0)	0.09, 0.157 [f]

LRP = low risk perception group, HRP= high-risk perception group. Each item was checked or not, the percentage for checked items is presented, differences were determined using Chi-square tests or Fisher's exact tests indicated by [f]. Effect size was determined using Cramer's V. Significance was set at $p \leqslant 0.05$.

4. Discussion

This interdisciplinary study combined social science with performance and hydration physiology to investigate how RP relates to physiological strain, subjective responses, and behavioral strategies in hikers exposed to hot and dry environmental conditions. Three hypotheses were formulated. H1 predicted that self-reported perceived exertion, heat stress, and thirst after a hike would be higher in individuals with high RP scores. The results partially supported this hypothesis: hikers with higher RP reported greater heat stress and thirst, but not greater perceived exertion. H2 proposed that physiological markers, such as elevated core body temperature, high urine concentration, and low fitness level, would be associated with lower RP. This hypothesis was not supported, as these physiological outcomes were unrelated to RP. H3 predicted that familiarity with heat risks and mitigation strategies, including preparation for heat and the use of hydration strategies, would be greater in hikers with higher RP. The results did not support the first two components of H3, as familiarity with heat warnings, the 'Take a Hike, Do it Right' campaign, and general preparation behaviors were not significantly different across RP groups. However, the third component of H3 was supported: hikers with high RP reported giving more weight to planned hike duration and daily ambient temperature when planning fluid intake.

4.1. RP

RP has been described in prior work in different ways [34]. Here, the RP score (48 out of 80 points) was mainly constructed from questions about worry [23], reflecting uncertainty about particular situations [34]. The current study revealed that physically-fit and well-informed hikers worried about accidents when walking on a hard trail, insufficient and inadequate peers, exertional heat illness, physical exhaustion, and a lack of shade while hiking on hard trails in a hot and dry environment. Aside from the present study, only one other investigation in China [23] examined RP among hikers; conducted in a

non-desert environment and, found that hikers were primarily concerned with sudden weather changes, security issues and traffic accidents. Our study contributes to a greater understanding of RP among hikers, especially in a desert environment.

It has been suggested that when individuals are exposed to heat, their perceptions of risk related to health and safety may decrease, thereby making them more likely to engage in risky behaviors [16]. At the same time, it has been reported that hikers with higher RP were more likely to have engaged in certain protective behaviors [35]. While it has been reported that hikers with high-risk more frequently report checking the weather conditions before their hike, no difference was reported for the amount of water carried [35]. This is in contrast with our study, as we show two-thirds of participants considered the planned duration of the hike and the ambient temperature while planning how much fluid to bring, and in all cases, the HRP group reported these behaviors more frequently than the LRP group. Some hikers may worry more than others, but these differences do not necessarily lead to different practical outcomes, for example, for the amount of fluid brought or consumed. Further, RP could be shaped by sex or gender and socioeconomic factors, and for example Huang *et al* have shown that RP can be shifted by an intervention [36], which is interesting as it may help to safeguard hikers at risk.

4.2. Self-reported perceived exertion, heat stress, thirst, and RP

The HRP group reported higher self-reported scores for heat stress and thirst. For thirst, this was specifically reflected in a higher 75th percentile score compared to the low RP group. This suggests a possible relationship between perceiving higher levels of risk and physical manifestations of heat and thirst, which is in line with the general observation that heat exposure affects an individuals' RP [16]. In earlier research, low perceptions of risk were associated with a faster starting pace [37], but in the current study, no difference was seen for the rate of perceived exertion (Borg scale) between RP groups, while rate of perceived exertion directly positively relates to work rate and pacing [38].

4.3. Physiological metrics, self-reported outcomes, and RP

The RP score, based on perceived dangers and worries about the extreme Phoenix heat, was not significantly associated with the post-hike collected physiological test outcomes, such as fitness level, body temperature estimates via thermal imaging (head shot), and hydration (i.e. urine concentration (USG), or fluid carried and consumed).

The estimated VO_2max in the current study, measured through conducting a submaximal stepping test in a subsample (table 3), was above average to excellent ($\geqslant$41.5 ml kg $^{-1}$min^{-1} for 40–49 year old males) [39]. It is important to emphasize that most of the hikers were recruited at the Piestewa Peak trail, rated extremely difficult by the City of Phoenix. It covers a 365 m (1,200-foot) elevation gain within 1.7 km (1.05 miles), i.e. a 21.5% grade, and provides almost no shade cover. Interestingly, the current study showed slightly higher numerical VO_2max values for the low RP group vs. the high RP group, suggesting that fitness level may be related to RP. Although no studies directly link VO_2max (a measure of cardiorespiratory fitness) to RP, individuals with higher hiking frequency (potentially indicative of greater fitness levels) perceived lower risks [23]. Previous research from Phoenix also showed that less fit individuals experienced more physiological complaints during a hiking activity in the heat than their fitter counterparts [40].

The estimation of body temperature via infrared thermal imaging was selected as a non-invasive measurement [33]. Its accuracy in determining skin temperature [41] and body core temperature has been debated [32]. Infrared thermal imaging has been suggested to be of some practical relevance for screening larger athletic groups [32, 42]. In contrast to an earlier study indicating that thermal imaging can be used to quantify body temperature changes [43], the current study showed no correlation with any of the other physiological parameters, and no differences in temperatures were seen between HRP and LRP groups. In addition, the measured body temperature itself was in the range (35 °C–36 °C) of previously reported skin temperature values in hiking conditions reported by our lab [44, 45], as such the suitability of the infrared thermal imaging vs. more precise methods to estimate body core temperature could be questioned.

The hydration state based on USG was similar to earlier reporting in hikers [40, 46, 47]. In theory, cognitive function is associated with hydration status, and in the case of dehydration, decrements in cognitive function could affect mood state [48]. At the same time, it has been shown that mood state can increase RP [49]. However, the current study results did not show any relation between hydration outcomes (i.e. USG, fluid brought, and fluid consumed) and RP.

Overall, it is unclear why RP did not stronger aligned with these physiological outcomes. Each of the measurements itself had its limitations, for example the YMCA step test is a submaximal test, which

includes several assumptions, while it was performed after a hike, whereas someone normally would perform this standardized test in a rested state [27].

4.4. Strengths and limitations

This study combined questionnaire-based responses with actual physiological measurements directly after a hike. Most of the data was collected at a difficult trail under hot conditions. The use of an exit questionnaire resulted in a substantially larger number of participants contributing to the physiological measurements compared to earlier studies that included pre- and post-hiking measurements [40, 46]. Limitations included the use of questions to determine RP, that although used in previous studies and exhibiting internal reliability have not yet been subjected to any comprehensive psychometric validation. Fitness was based on standardized submaximal fitness test that took place after the hike, which may have impacted the accuracy of the estimated fitness level. The use of non-invasive thermal imaging to assess body temperature likely resulted in a less accurate outcome than using (continuous) core body temperature sensors. A limitation of this study is that most participants were recruited from an 'extremely difficult' trail, which may have biased the sample toward individuals with higher fitness levels. The small supplemental sample recruited from less active participants performing on an easier trail in Year 2 ($n = 21$) was not sufficient to fully balance this, and thus the findings should be interpreted with some caution regarding their generalizability. Finally, only those who hiked for more than 30 min were included in the data analysis.

5. Conclusion

The collective findings suggest that RP is linked to subjective experiences of heat stress and to selected hydration-related behaviors, but not to physiological strain or general familiarity with heat risk. These findings underscore the complexity of how individuals perceive and respond to environmental risks in heat-exposed recreational settings. Our results have important practical implications:

First, the fact that high RP correlates with subjective discomfort (heat and thirst) but not with physiological strain suggests that RP may be more strongly tied to perceptual and cognitive factors than to actual physical danger in this context. This disconnection implies that interventions focused solely on disseminating physiological facts about hydration and heat risk may not fully influence how people perceive or prepare for heat. Instead, it is suggested to investigate in future research the impact of tailored education campaigns that also address the psychological aspects of risk, such as anxiety about accidents or discomfort from heat.

Second, most participants considered ambient temperature and hike duration when deciding how much water to bring, yet this did not result in differences in hydration status. This result highlights a gap between planning and execution and suggests that even when people are aware of risk factors and take preparatory actions, their hydration strategies may still be insufficient or not well-aligned with actual physiological needs. This is a critical insight for park managers and public health officials: current messaging, such as the 'Take a Hike, Do it Right' campaign, may raise awareness but does not necessarily lead to behavior changes that improve physiological safety outcomes.

Based on these conclusions, our study emphasizes the need for more dynamic, real-time interventions. Therefore, future research should focus on the efficacy of the use of technologies, such as mobile alerts, that communicate weather risks or hydration reminders based on exertion that could bridge the gap between awareness and effective action. Similarly, it is worthwhile to investigate if making hydration stations or shaded rest stops more visible and accessible could reduce the reliance on self-assessment, which our study shows is imperfect.

Finally, our study contributes to the understanding of how individuals navigate environmental risks under climate stress. As heat extremes become more frequent due to climate change, understanding the psychological and behavioral components of heat risk becomes increasingly important, not only for recreationists but also for outdoor workers and other vulnerable populations. Future research should explore whether similar patterns exist in different populations (e.g. less fit hikers, tourists unfamiliar with desert conditions, or occupational groups), and whether real-time physiological feedback can improve safety outcomes more effectively than traditional education campaigns.

We demonstrated that RP is an essential but imperfect guide for behavior in extreme heat. Interventions that consider both the psychological and physiological dimensions of heat risk are likely needed to better safeguard individuals engaging in outdoor activities in hot climates.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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